

GRADUATE STUDENT HANDBOOK

Department of Civil, Construction and Environmental Engineering
Iowa State University

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1. INTRODUCTION

The Department of Civil, Construction and Environmental Engineering (CCEE) *Graduate Student Handbook* is intended to provide academic guidance for students enrolled in the CCEE graduate programs. This document references key resources in the Iowa State University (ISU) and CCEE Department, identifies degree components and requirements, and explains key steps towards graduation for students pursuing a graduate degree in civil engineering. The handbook does not contain all the resources and policies that may be relevant, but it helps direct the students to obtain additional information.

The [ISU Graduate College Handbook](#) is a comprehensive resource for policies and procedures that pertain to graduate students. The [Graduate College](#) posts the Graduate Handbook, as well as many important resources, including academic milestones, degree requirements, graduate financing, thesis/dissertation expectations, and deadlines associated with graduation. All graduate students are expected to make themselves aware of both *ISU Graduate College Handbook* and *CCEE Graduate Student Handbook*.

The CCEE Graduate Student Handbook is consistent with the policies and procedures formulated by ISU and the Graduate College and, hence, supplements the information provided by them. This handbook is intended to emphasize additional departmental requirements and resources. Note that some specialization areas within the department may have further requirements regarding core courses, graduate seminars, meetings, or other activities.

EACH STUDENT IS RESPONSIBLE FOR MEETING ALL POLICIES, DEADLINES, AND REQUIREMENTS.

This handbook can be found on the CCEE website.

2. DEGREE PROGRAMS

2.1 Graduate Degree Major

- A. The CCEE Department offers the following degree programs:
 - Master of Engineering (M.Eng.): coursework only
 - Master of Science (M.S.): thesis or creative component
 - Doctor of Philosophy (Ph.D.)
- B. The degree major is **Civil Engineering** with areas of specialization in:
 - Civil Engineering Materials
 - Construction Engineering and Management
 - Environmental Engineering
 - Geotechnical Engineering
 - Structural Engineering
 - Transportation Engineering
 - Transdisciplinary Infrastructure Engineering (M.S. and Ph.D. only)
 - General Civil Engineering (M.Eng. only)
- C. For all degree programs, the diploma will list **Civil Engineering** as the major.
- D. M.Eng. students have the option of earning a degree in **Civil Engineering**, without a specialization. This *General Civil Engineering* option is only available for the students with an accredited bachelor's (B.S.) degree in civil, construction, and environmental engineering.

2.2 Interdepartmental Degree Programs

- A. The CCEE department participates in the following interdepartmental degree programs:
 - [Environmental Science](#): M.S. and Ph.D.
 - [Interdepartmental Microbiology Graduate Program](#): M.S. and Ph.D.
 - [Wind Energy Science, Engineering & Policy \(WESEP\)](#): Ph.D. only
- B. Interdepartmental program requirements are established by the supervisory committee of each program. Those requirements are in addition to the requirements set for the CCEE graduate degree programs.

- C. For additional information on policies regarding interdepartmental programs, refer to the [Graduate College Handbook, Section 4.1.3.](#)

2.3 CCEE B.S./M.S. Concurrent Program

- A. Eligible CCEE undergraduate students in civil, construction, and environmental engineering can become graduate students *before* they graduate with their B.S. degree.
- B. CCEE juniors and seniors are encouraged to consult the [CCEE Undergraduate Advising Center](#) and the CCEE Graduate Admissions Coordinator regarding B.S./M.S. concurrent program eligibility requirements.
- C. If accepted, the student's status changes from undergraduate to graduate during the concurrent period, and all tuition and fees will be assessed as a graduate student.
- D. After the B.S. degree is complete, concurrent students continue to finish their M.S. degree program.
- E. Up to 6 credits of graduate-level formal courses completed during the concurrent period can be counted for both the B.S. and M.S. degrees.
- F. For additional information on concurrent programs, refer to the [Graduate College Handbook, Section 4.3.2.](#)

2.4 CCEE Graduate Courses Taken as an Undergraduate

- A. If students choose to start a master's degree *after* they graduate with their B.S. degree, they can count up to 9 credits of graduate-level formal courses completed during their undergraduate program.
- Six credits can be counted for both B.S. and master's degree requirements.
 - Three credits can be counted for master's degree requirements only if they are not included in B.S. degree requirements (i.e., completed in addition to B.S. degree requirements).
- B. For additional information, refer to the [Graduate College Handbook, Section 6.3.2.](#)

2.5 Graduate Course Decisions

- A. Undergraduate students interested in graduate degree programs should take their specialization area of interest into account, so that the courses completed can count for the graduate program.

3. GENERAL PROGRAM REQUIREMENTS

3.1 Program Requirements

- A. The details of each graduate student's program are developed in consultation with a Major Professor and an Academic Plan Committee (APC). Appendix A provides a list of required courses for each specialization area.
- B. Total credit requirements
- 30 credits for M.S. and M.Eng.
 - 72 credits for Ph.D.
- C. M.S. options:
- Thesis
 - Creative Component

3.2 Prerequisite Course Requirements

- A. The courses listed in Appendix B are **minimum undergraduate course requirements** for students seeking a graduate degree in the areas of specialization in the CCEE Department.
- Equivalent courses may be substituted for the listed courses.
 - Students who are missing prerequisite courses can be admitted with the requirement that they complete prerequisite courses on a graded basis before attempting graduate-level formal courses.
 - Students are expected to consult their Major Professor regarding any additional prerequisite course requirements based on their areas of specialization. Prerequisite courses do not count towards the graduate degree.
- B. Students *without* an accredited B.S. degree in engineering should be aware that completion of the graduate degree may *not* qualify them to attain registration as a professional engineer in some states. The licensing board in the

state the student desires registration should be consulted for information about current and developing policies regarding the qualifications for engineering licensure.

3.3 Thesis, Creative Component, and Dissertation Definition & Requirements

- A. **Master's thesis**, as defined by the [Graduate College Handbook, Section 7.1.4](#):
 "A master's thesis is a scholarly composition that demonstrates the ability of the author to do independent and creative work. It explores in some depth a problem or issue related to the major field of study. Although considerable variations in format and style are acceptable, precise expression, logical construction, and meticulous attention to detail are essential."
- B. **Doctoral dissertation**, as defined by the [Graduate College Handbook, Section 7.1.4](#):
 "A doctoral dissertation must demonstrate conclusively the ability of the author to conceive, design, conduct, and interpret independent, original, and creative research. It must attempt to describe significant original contributions to the advancement of knowledge and must demonstrate the ability to organize, analyze, and interpret data. In most instances, a dissertation includes a statement of purpose, a review of pertinent literature, a presentation of methodology and results obtained, and a critical interpretation of conclusions in relation to the findings of others. When appropriate, it involves a defense of objectives, design, and analytical procedures. Dissertation research should be worthy of publication and should appear in appropriate professional journals or in book form."
- C. **Master's creative component**, as defined by the [Graduate College Handbook, Section 7.1.5](#):
 "Non-thesis students must present substantial evidence of individual accomplishment (e.g., a special report, capstone course, integrated field experience, annotated bibliography, research project, design, or other creative endeavor)."
- D. For theses and dissertations, the [Graduate College Handbook, Section 7.1.4](#) states:
 "Since satisfactory completion of the dissertation or thesis can constitute one of the most gratifying experiences in graduate study, the document should reflect the highest standards of scholarship, serving as a measure of quality for the student, Major Professor, the program, and Iowa State University."

3.4 Graduate English Requirements for Non-Native Speakers of English

- A. Graduate students whose native language is not English must follow ISU Admissions and Graduate College policies for English proficiency.
- B. The CCEE Department does not require the Graduate Records Examination (GRE). However, GRE scores can be reported as a measure of evaluation during the application review process.
- C. **Prior to Admission:** English proficiency examination
 - This examination requirement measures an applicant's understanding of the English language.
 - Applicants are exempt from the English proficiency examination if they have earned a bachelor's or higher degree from an English-speaking country, following the details in [Graduate Application Information](#).
- D. Applicants who do not meet the exemption criteria must take one of the following exams:
 - [TOEFL: Test of English as a Foreign Language](#)
 - [IELTS: International English Language Testing System](#)
 - [Pearson Test of English \(PTE\)](#)

The ISU Office of Admissions and CCEE Department minimum English requirements are listed as follows:

Exam	Admissions Minimum	CCEE Minimum
TOEFL iBT	79	82
IELTS	6.5	6.5
PTE	53	55
Duolingo	115	115

- E. **After Admission:** English Placement Test (EPT)
 - The [EPT](#) administered by ISU is a measure of a student's ability to communicate using the English language.

- Students are exempt from the EPT if they achieve or exceed scores on the following exams:

Exam	Score meets or exceeds
TOEFL iBT	100
IELTS	7.5
PTE	72

- Graduate students should take the EPT as soon as they begin their graduate program, preferably before the start of classes in their first semester, but no later than the end of the first semester.
- The EPT is pass/fail and is scored by staff in the ISU English Department.
- A student who passes the EPT fulfills ISU English proficiency requirements.
- A student who does not pass the EPT is assigned to English courses, which must be completed during the first year of study.
- For additional information, refer to the [EPT](#) and the [Graduate College Handbook, Section 4.4.3](#).

4. GRADUATE DEGREE OPTIONS AND CREDIT REQUIREMENTS

4.1 Guidelines for All Degree Programs

- According to the [Graduate College Handbook, Section 5.1.1](#), graduate courses are taught by members of the graduate faculty and are given numbers in the 5000s or 6000s. For the purpose of *CCEE Graduate Student Handbook*, they are referred to as **formal courses** and are **separate from 5900, 5990, 6990**, and similar research-based courses.
- According to the [Graduate College Handbook, Section 5.1.4](#), credits for a special report, research project, or other independent work designated as a creative component under the nonthesis master's degree option are assigned under the course number **5990**. These credits are not calculated in the student's GPA.
- According to the [Graduate College Handbook, Section 5.1.8](#), credits for research for the purpose of completing a dissertation or thesis are assigned under the course number **6990**. These credits are not calculated in the student's GPA.
- According to the [Graduate College Handbook, Section 5.1.10](#), credits for independent study with a faculty member by an individual student or for a small group studying a specialized area are assigned under the course number **5900**. These credits are calculated in the student's GPA.

4.2 Master of Science

- All guidelines in Section 3.
- The student must select an area of specialization within the CCEE Department.
- Minimum of **30 total credits**, with the following conditions:
 - Minimum** of 22 credits of graduate-level formal courses for the thesis track and 27 credits of graduate-level formal courses for the creative component track
 - Minimum** of 20 credits of graduate-level formal courses and CE 5990 or CE6990 in the area of specialization
 - Minimum** of 6 credits of graduate-level formal courses outside of the area of specialization (as approved by APC)
 - Minimum** of 6 credits of CE 6990 (for thesis track)
 - Minimum** of 2 credits of CE 5990 (for creative component track)
- Additional considerations for the Academic Plan:
 - Maximum** of 3 credits of 4000-level Civil Engineering (CE) formal courses are allowed (see Table 1 for the list of the allowed courses). This must be outside the area of specialization.
 - Maximum** of 3 credits of CE 5900 (Independent Study) are allowed.
 - Maximum** of 3 credits of Graduate Studies (GRST) courses are allowed towards meeting outside the area of specialization coursework requirements.

- **Maximum** of 9 credits of CE graduate-level formal courses completed as an undergraduate student (refer to Section 2.4) are allowed.
- **Maximum** of 9 credits of CE graduate-level formal courses completed as an ISU nondegree graduate student are allowed.

4.3 Master of Engineering

- A. All guidelines in Section 3.
- B. The student may select an area of specialization within the CCEE Department or choose to proceed with General Civil Engineering.
- C. Minimum of **30 total credits**, with the following conditions:
 - **Minimum** of 18 credits of graduate-level formal courses in the area of specialization. For General Civil Engineering, the graduate-level formal courses can be from up to a maximum of 3 areas of specialization within the CCEE Department, as approved by Major Professor.
 - **Minimum** of 6 credits of graduate-level formal courses outside of the area of specialization or CCEE Department (as approved by Major Professor)
- D. Additional considerations for the Academic Plan:
 - **Maximum** of 3 credits of 4000-level Civil Engineering (CE) formal courses are allowed (see Table 1 for the list of the allowed courses). This must be outside the area of specialization.
 - **Maximum** of 3 credits of 5900 (Independent Study) are allowed. This is counted as a graduate-level formal course (beyond the minimum credits required for in and outside the area of specialization).
 - **Maximum** of 9 credits of CE graduate-level formal courses completed as an undergraduate student (refer to Section 2.4) are allowed.
 - **Maximum** of 9 credits of CE graduate-level formal courses completed as an ISU nondegree graduate student are allowed.

4.4 Doctor of Philosophy

- A. All guidelines in Section 3.
- B. The student must select an area of specialization within the CCEE Department.
- C. Minimum of **72 total credits**, with the following conditions:
 - **Minimum** of 33 credits of graduate-level formal courses
 - **Minimum** of 12 credits of graduate-level formal courses in the area of specialization*
 - **Minimum** of 9 credits graduate-level formal courses outside the CCEE Department (as approved by APC)
 - **Minimum** of 16 credits of CE 6990
- D. Additional considerations for the Academic Plan:
 - **Maximum** of 6 credits of 4000-level Civil Engineering (CE) formal courses are allowed (see Table 1 for the list of the allowed courses). This must be outside the area of specialization.
 - **Maximum** of 6 credits of 5900 (Independent Study) are allowed. This is counted as a graduate-level formal course (beyond the minimum credits required for in and outside the area of specialization).
 - **Maximum** of 3 credits of Graduate Studies (GRST) courses are allowed towards meeting outside the CCEE Department coursework requirements.
 - **Maximum** of 9 credits of CE graduate-level formal courses completed as an undergraduate student (refer to Section 2.4) are allowed.
 - **Maximum** of 9 credits of CE graduate-level formal courses completed as an ISU nondegree graduate student are allowed.

*For Transdisciplinary Infrastructure Engineering, the graduate-level formal courses selected to meet this 12-credit requirement can be from any conventional areas of specialization within the CCEE Department (as approved by APC).

Table 1. CCEE 4000-Level Courses Approved for Graduate Credit Outside Specialization Area and Minor Graduate Credit

Specialization Area	CCEE Courses Permitted
Civil Engineering Materials	CE 4170, CE 4280, CE 4460, CE 4480
Construction Engineering and Management	CE 4170, CE 4280, CE 4460, CE 4480, CE 4600
Environmental Engineering	CE 4170, CE 4460, CE 4480, CE 4600
Geotechnical Engineering	CE 4170, CE 4280, CE 4460, CE 4480
Structural Engineering	CE 4170, CE 4280, CE 4600, CE 4670, CE 4840
Transportation Engineering	CE 4170, CE 4280, CE 4460, CE 4480, CE 4600
Transdisciplinary Infrastructure Engineering	CE 4170, CE 4280, CE 4460, CE 4480, CE 4600, CE 4670, CE 4840

4.5 Transfer Credits of Graduate-Level Formal Courses

- Graduate-level formal course credits received from ISU for the M.S. degree can all be included as part of a Ph.D. Academic Plan, subject to the approval of the APC.
- Graduate-level formal course credits completed at other institutions may be used towards the Ph.D., M.S., or M.Eng. degree program at ISU. However, they are subjected to the review and approval of the APC, CCEE Director of Graduate Education (DOGE), and the Graduate College.
- The transfer credits can be CE and non-CE graduate-level formal courses, excluding research and professional development courses. The credits used for earning a graduate-level degree outside ISU cannot be used again towards a similar-level degree at ISU.
- The transfer credits must have been completed as a graduate student, and the grade received must be equivalent to the ISU letter grade of B or better.
- The transfer credits completed at other institutions are limited to 12 credits towards meeting the Ph.D. coursework requirements and 6 credits towards meeting the M.S. or M.Eng. coursework requirements.

4.6 Co-Specialization Options (M.S. and Ph.D. in Civil Engineering)

- The CCEE Department permits the pursuit of a co-specialization option if both specializations are approved by the student's APC.
- The desire for such a program would result from a research and/or professional interest that does not fall within a single specialization area but is within the scope of two areas.
- The degree is to be granted when the student fulfills separately the requirements of each specialization.
 - For an M.S. degree with co-specialization, **minimum** of 9 graduate-level formal course credits are required in each of the two areas of specialization
 - For a Ph.D. degree with co-specialization, **minimum** of 12 graduate-level formal course credits are required in each of the two areas of specialization.
 - The other credit requirements remain unchanged. Examinations and research deliverables associated with a co-specialization option should be related to both specializations. The student's APC is expected to have a Co-Major Professor from each of the two areas of specialization.

4.7 Co-Major Degree Program

- A co-major is a program of study for a single degree in which the requirements for two separate majors are met.
- An M.S. degree student, outside the CCEE department who wishes to co-major in an area of specialization within Civil Engineering, is required to take a **minimum** of 12 graduate-level formal course credits in that area of specialization. The other credit requirements remain unchanged.
- A Ph.D. degree student, outside the CCEE department who wishes to co-major in an area of specialization in Civil Engineering, is required to take a **minimum** of 12 graduate-level formal course credits in that area of specialization. The other credit requirements remain unchanged.

- D. Students must meet all degree requirements set by both major degree programs. The student's APC is expected to have Co-Major Professors, one from the CCEE Department and one from the student's other co-major department.
- E. Examinations and research deliverables associated with a co-major option should be related to both major degree programs.
- F. Students without a civil engineering bachelor's degree may pursue a co-major in a civil engineering area of specialization if appropriate background courses are successfully completed. The minimum undergraduate course requirements for each area of specialization are listed in Appendix B.
- G. CCEE students who wish to co-major in another degree program should consult the DOGE of that degree program with regards to the co-major degree requirements.
- H. For additional information, refer to the [Graduate College Handbook, Section 4.3.1](#).

4.8 Minor for Non-Civil Engineering M.S. Major Degrees

- A. **Minimum** of 9 credits of graduate-level formal courses are required in the specialization area selected within the CCEE Department.
- B. One graduate faculty member from the specialization area selected within the CCEE Department must be the minor representative/committee member on the APC.
- C. The minor committee member will guide the student in specific course requirements and administer written or oral questions during the final oral exam.

4.9 Minor for Non-Civil Engineering Ph.D. Major Degrees

- A. **Minimum** of 12 credits of graduate-level formal courses are required in the specialization area selected within the CCEE Department.
- B. One graduate faculty member from the specialization area selected within the CCEE Department must be the minor representative/committee member on the APC.
- C. The minor committee member will guide the student in specific course requirements and administer written or oral questions during the preliminary and final oral exams.

5. NON-DEGREE PROGRAMS

5.1 Non-Degree Program – General

- A. Qualified students may enroll in a non-degree graduate program with a declared area of specialization in Civil Engineering.
- B. These programs are intended to provide advanced study opportunities to students who wish to increase their technical competence, but who are not interested in pursuing the entire graduate degree program.
- C. A **maximum** of 9 graduate-level formal course credits earned under the non-degree option may be applied to a graduate degree if the student later chooses to undertake a degree option.
- D. A Major Professor from the CCEE Department will be appointed to advise the student.

5.2 Environmental Engineering and Environmental Systems Graduate Certificates

- A. Students must have earned an undergraduate engineering degree or have met the prerequisite undergraduate courses shown in Appendix B for the Environmental Engineering specialization.
- B. Environmental Engineering and Environmental Systems Graduate Certificate require a **minimum** of 12 approved graduate-level formal course credits.
- C. Students should check with their Major Professor or the Environmental Engineering graduate coordinator for additional guidance.

5.3 Construction Management Graduate Certificate

- A. Construction Management Graduate Certificate requires a **minimum** of 12 approved graduate-level formal course credits. The selected courses must include:
 - CE 5010 (3 credits): Preconstruction Project Engineering and Management

- CE 5020 (3 credits): Construction Project Engineering and Management
 - CE 5030 (3 credits): Construction Management Function and Processes
- B. Students should check with the Construction Management Director of Certificate Studies for additional guidance.

6. DEVELOPING THE PROGRAM OF STUDY

6.1 Major Professor

- A. New incoming graduate students are required to arrange to meet with their Major Professor within the first two weeks of their first semester.
- B. If there is a student without a Major Professor, the CCEE DOGE will act as temporary Major Professor and work with the student to identify a permanent Major Professor.
- C. Table 2 provides guidance on Major Professor selection.

Table 2. Major Professor Selection

Student	Major Professor
With research assistantship	Faculty member who provides the research assistantship
With teaching assistantship	Faculty member who had indicated willingness to work with the student during the admission process or a professor identified by the student's specialization area graduate coordinator serving as temporary Major Professor until a permanent Major Professor is identified
Self-funded students (M.S. or Ph.D.)	Faculty member who had indicated willingness to work with the student during the admission process or a professor identified by the student's specialization area graduate coordinator serving as temporary Major Professor until a permanent Major Professor is identified
M.Eng. students	Identified and assigned by the specialization area graduate coordinator
Certificate students	Director of Certificate Studies (DOCS) for the pursued certificate

6.2 Responsibilities of the Major Professor

- A. Direct the student's graduate study and serve as the chair of the APC
- B. Guide the student in the selection of courses and the development of the Academic Plan
- C. Assist the student in defining educational and professional objectives and developing a program of study to meet the set objectives
- D. Supervise the student's research work, thesis, or creative component
- E. For additional information about the Major Professor, refer to the [Graduate College Handbook, Section 6.4.1.](#)

6.3 Academic Plan Committee (APC)

- A. The student and Major Professor should discuss and identify possible committee members. The APC must be processed and approved before the Academic Plan can be submitted.
- B. Possible committee members should have relevant expertise to contribute to the student's research or creative component work.
- C. All prospective committee members should be contacted personally by the student and asked to serve in this capacity.
- D. The **timeline** recommended for forming an APC is as follows:
 - M.S. and M.Eng. students: Before the end of the first semester
 - PhD students: Before the end of the first year
- E. Table 3 outlines the minimum requirements for APC membership.

Table 3. Academic Plan Committee (APC) Membership

Degree	Committee Membership
Master of Science (M.S.)*	• Minimum of three members of the graduate faculty • Minimum of two members from the area of specialization • Minimum of one member from outside the area of specialization. This member can be a CCEE faculty member from a different specialization area or from outside the CCEE Department.
Doctor of Philosophy (Ph.D.)*	• Minimum of five members of the graduate faculty • Minimum of three members from the area of specialization • Minimum of two members from outside the area of specialization • Minimum of one member from outside the CCEE Department
Master of Engineering (M.Eng.)	• Major Professor only
Certificate	• Director of Certificate Studies (DOCS)

- F. In situations where there are insufficient faculty in an area of specialization, committee members can be from a related area of specialization. It can be to the student's advantage to include additional committee members to broaden the perspective for the student's program of study.
- G. In consultation with the Major Professor, the student should select APC members who can make meaningful contributions to the student's educational and research programs.
- H. If a graduate minor has been declared, a graduate faculty member from the minor program or interdepartmental minor must serve on the APC. The Major Professor and the representative from the minor cannot be the same person, following the [Graduate College Handbook, Section 4.1.4](#).
- I. The student should seek guidance from the Major Professor and APC members during the course of the graduate study, and not just at the time of oral examinations.
- J. Meetings of all members of the APC should be held as frequently as necessary to guide the student's program of study and to evaluate progress.
- K. As a **minimum**, formal meetings of the APC should be held:
 - To approve the Academic Plan, including research or creative component topic.
 - For Ph.D. students, all APC members must participate in the preliminary oral exam.
 - For Ph.D. and M.S. students, all APC members must participate in the final oral exam.
- L. For additional information about the APC, refer to the [Graduate College Handbook, Sections 6.1 and 6.2](#).

6.4 Academic Plan

- A. The student and the Major Professor develop the Academic Plan with the consultation and approval of the APC.
- B. Each student's program of study should be:
 - Consistent with the CCEE degree and credit requirements (Section 4).
 - Designed to meet prerequisite requirements for the target CCEE degree and specialization area.
 - Planned based on the student's research and professional interests while meeting program requirements.
 - Tailored to result in scholarly products (for M.S. and Ph.D. students), following [authorship policies](#).
- C. The Academic Plan Committee ensures that degree and credit requirements are met before approving the Academic Plan.
- D. To process the Academic Plan, students should enter only those courses that are necessary to meet degree and credit requirements.
 - English courses required for international students should not be included.
 - Any courses taken for the student's preference, but not applicable to the program of study, should not be included (e.g., music, swimming, etc.).
 - Any courses completed, but not on the Academic Plan, will be shown on transcripts.

- E. Once the Academic Plan is submitted, it is approved in the following order: Graduate College > Academic Plan Committee > CCEE DOGE. Students should plan for at least a month to get the Academic Plan fully approved. The Academic Plan must be approved according to the [Graduate College graduation deadlines](#).
- F. While it is to the best of the student's interest to submit their Academic Plan as soon as they know what courses they will take, the Academic Plan has to be approved **no later than**:
 - M.Eng. and M.S. students: The end of the semester before the semester a student intends to graduate
 - Ph.D. students: 3 months prior to the preliminary oral examination

6.5 Credit Load

- A. CCEE Department follows credit load limits as listed in the [Graduate College Handbook, Section 2.1.2](#).
- B. Full-time students with research/teaching assistantships
 - Graduate assistants during the academic year are considered full time. Refer to the [Graduate College Handbook, Section 2.1.1](#). For their assistantship to be processed, enrollment in a minimum of one credit is required.
 - For quarter-time appointment: Maximum of 15 credits for fall/spring and 10 credits for summer
 - For half-time appointments: Maximum of 12 credit for fall/spring and 6 credits for summer
- C. Full-time students sponsored externally by a foundation, a government agency, or another university
 - Minimum of 9 to a maximum of 15 credits for fall/spring
 - No minimum but a maximum of 10 credits for summer
 - When preparing the Academic Plan, the student and the Major Professor are obligated to meet objectives set forth in the sponsoring agency's program.
- D. Full-time students with self-support
 - Minimum of 9 to a maximum of 15 credits for fall/spring
 - No minimum but a maximum of 10 credits for summer
 - The student is responsible to their Major Professor for fulfilling program requirements and for participating in activities designed for their academic and/or professional development.
- E. Continuous Enrollment Requirement
 - Ph.D. students who have 1) passed the preliminary oral examination, and 2) fulfilled Academic Plan requirements of coursework and research, **but have not** yet completed the final oral examination, **must register** for a minimum of 1 credit during spring and fall semesters. In consultation with the Major Professor, this can be one credit of CE 6990: Research or GRST 6810A: Doctoral Continuous Registration. Tuition and fees apply.
 - Refer to the [Graduate College Handbook, Section 2.2.2](#) for additional information.
- F. Examination Registration
 - Ph.D. students must be registered for a **minimum** of 1 credit in the term of the preliminary oral examination.
 - Ph.D. and M.S. students must be registered for a **minimum** of 1 credit in the term of the final oral examination.
 - If all Academic Plan courses and research credits have been completed, students can register for 1 credit of GRST 6810B: Examination Only. Tuition and fees apply.
 - If the final oral exam takes place in between semesters, students can register either the semester before or the semester after the exam.
 - Refer to the [Graduate College Handbook, Section 2.2.1](#) for additional information.

7. ACADEMIC PERFORMANCE AND PROGRESS

7.1 Graduate Student Rights and Responsibilities

- A. Graduate students are expected to demonstrate a commitment to their academic endeavors, to make steady progress toward academic milestones and goals, and to demonstrate ethical practices at all times.
- B. The following policies and laws apply to graduate students or assistants: *Student Code of Conduct, Academic Misconduct, Discrimination and Harassment Policy, Sexual Misconduct, Sexual Assault, Sexual Harassment, Stalking, and Intimate Partner Violence Involving Students, Conflicts of Interest, Statement on Professional Ethics, Compliance*

with Research Standards, and Ownership of Intellectual Property and Data. For detailed information, refer to the [Graduate College Handbook, Section 8](#).

7.2 Time-to-Degree Limit

- A. The Graduate College allows all graduate students, Ph.D., M.S., and M.Eng., 7 years to complete their degrees.
- B. For additional information, refer to the [Graduate College Handbook, Section 4.4.4](#).

7.3 GPA Requirement

- A. All graduate students are required to maintain a cumulative grade point average (GPA) of 3.0 to remain in satisfactory standing.
- B. The GPA is calculated using the standard point allocation for university grades.
- C. All courses taken as a graduate student will be calculated into the graduate GPA, even if they are not included on the Academic Plan. Examples include undergraduate courses outside those specified by the program, additional courses of interest, and English courses required for international students.
- D. Grades for program requirements that carry R (Required) or 0 credit are not calculated into the graduate GPA. However, any grade of F (Fail) or I (Incomplete) will prevent a student from graduating.
- E. Grades of I, S, P, NP, T, and W are not counted in the graduate GPA.
- F. Grades of F, even if taken S/F, are counted in the graduate GPA.
- G. The grade given for an I (Incomplete) is calculated into the graduate GPA.
- H. In the case of repeated courses, only the grade achieved the last time the course is taken is counted in the graduate GPA.
- I. Grades for creative component (CE 5990) or research (CE 6990) are not included in the GPA calculation.
- J. For additional information, refer to the [Graduate College Handbook, Section 5.2.3](#).

7.4 Academic Probation

- A. Students must maintain a **minimum GPA of 3.0** for all courses on their Academic Plan.
- B. A minimum grade of C is required for each course included in the Academic Plan.
- C. Students with course grades of C-, D, or D- are allowed to graduate if their GPA is 3.0 or higher, but the courses cannot be on the Academic Plan.
- D. Students with course grades of I, NP, or F are not allowed to graduate, even if their GPA is 3.0 or higher, regardless if they are included on the Academic Plan.
- E. Only in cases of extenuating circumstances are students with NP or F grades allowed to graduate. The course(s) in question cannot be related to the student's scholarly discipline and cannot be included on the Academic Plan. The Academic Plan Committee and CCEE DOGE must request special consideration for approval by the Dean of the Graduate College.
- F. Students not maintaining a GPA of 3.0 are placed on academic probation by the Dean of the Graduate College. The Graduate College places a hold on the student's registration for as long as the GPA is below 3.0.
- G. While on academic probation,
 - Students will not be able to register for classes without prior approval, as outlined in Section 7.4.H.
 - Ph.D. students will not be admitted to candidacy for a degree.
 - If appointed to a graduate assistantship, students will not receive a Graduate College tuition scholarship.
- H. Students on probation must be reviewed before registration for each term.
 - The Major Professor must review the student's record and make a recommendation to the CCEE DOGE regarding continued registration in the graduate program.
 - Upon review, the DOGE may approve and submit a Permission to Register request in Workday.
- I. For additional information, refer to the [Graduate College Handbook, Section 5.2.5](#).

7.5 Dismissal Criteria

- A. Failure to maintain academic standing
 - Inability to maintain a minimum GPA of 3.0 for two years
 - Failure to pass any required examinations within the time frame designated by the APC
 - Failure to complete any required coursework, thesis, or creative component credits within the time frame designated by the APC
 - Failure to demonstrate scholarly and professional competence
 - Failure to comply with the ethical standards of the university and profession
 - Academic probationary status for two years
 - Failure to establish a Major Professor within the first year of graduate study
 - Failure to comply with graduate student responsibilities or requirements discussed in the CCEE or Graduate College Handbooks.
 - For additional information, refer to the [Graduate College Handbook, Section 8.7](#).
- B. Academic/Research Misconduct
 - A code of conduct violation as determined by ISU Office of Student Conduct
 - For additional information, refer to [Academic Misconduct](#) and [Research Misconduct](#). Among the examples are the use of unauthorized sources, misrepresentation, falsification or fabrication of information, and plagiarism. Using artificial intelligence (AI) to generate academic work can be considered misconduct if it is used to plagiarize, cheat, or undermine the educational process.
- C. Non-Academic Misconduct
 - Conduct that violates Regents Uniform Rules of Personal Conduct and General University Regulations, as listed in the ISU Catalog and Policy Library: [Student Code of Conduct](#) and [Violence-Free University](#)

7.6 Dismissal Procedure

- A. A student's APC or, if the student has no APC, the student's (permanent or temporary) Major Professor, or the CCEE DOGE can recommend dismissal of a student for any of the dismissal reasons listed in Section 7.4.
- B. The dismissal procedure is as follows:
 - Informal conference: If the fitness of a graduate student to remain in the relevant academic program is questioned, an informal conference will be held with DOGE, the Major Professor, and the graduate student to attempt to resolve the matter.
 - Informal Conference with the Dean of the Graduate College is optional. If the situation cannot be resolved at the informal conference, either party may bring the problem to the attention of the Dean of the Graduate College. In attempting to resolve the matter, the Dean of the Graduate College will review the case, meet with the parties concerned, and attempt to identify alternatives to dismissal.
 - In cases where the steps outlined above do not lead to a resolution or acceptable improvement, the DOGE will notify the student in writing of dismissal. This notification will outline the reasons for dismissal and the effective date of the dismissal.
- C. For additional information, refer to the [Graduate College Handbook, Section 8.7](#).

7.7 Student Grievance Procedures

- A. Grades and Instruction
 - If a student feels their instructor has behaved unfairly or unprofessionally, a grievance must be invoked before a year passes after the course is completed.
 - If a student feels that they have received an unfair grade in a course, an appeal must be initiated by the midterm of the semester following the course completion.
 - In either case, the student is encouraged to contact their Major Professor, if the perceived violation does not involve the Major Professor.
 - If at all possible, academic grievances should be resolved with the instructor involved.

- If it is not possible to determine a resolution with the instructor, the student should submit in writing the grievance to the instructor's academic Department Chair.
 - If a resolution is not made with the Department Chair, the student can submit the grievance to the Dean of the instructor's college.
 - If there is no resolution with the Dean of the instructor's college, the student may forward the grievance to the Provost. An appeal of the decision of the Provost may be made to the President of ISU.
 - For additional information, refer to the [Graduate Student Handbook, Section 8.5.1](#).
 - Other sources of support outside the department include:
 - [Graduate College](#)
 - [Dean of Students office](#), Office of Student Assistance
 - [Ombuds Office](#)
 - [Graduate & Professional Student Senate \(GPSS\)](#)
- B. Scholarly and Professional Competence
- If a student feels their scholarly or professional competence has been evaluated unfairly, they should present their complaint to those most directly involved in the matter, which can be a faculty member, Major Professor, APC, DOGE, or Department Chair.
 - If a resolution is not possible with the individuals involved, the student can submit a grievance to the departmental grievance committee. If such a committee does not exist, the CCEE DOGE must appoint a committee.
 - If a student is unsatisfied with the findings of the departmental grievance committee, they may appeal the action in writing to the Dean of the Graduate College. The Graduate College grievance committee will then review the complaint.
 - Students submitting a grievance must be willing to participate in the grievance process.
 - Students still unsatisfied with the disposition of the grievance – on matters of procedure only – may appeal in writing to the Provost, and, if necessary, to the President of ISU.
 - For additional information, refer to the [Graduate Student Handbook, Section 8.5.2](#).
- C. Discrimination, harassment and/or sexual misconduct
- Report any concerns/incidents to the [Office of Equal Opportunity](#).

8. EXAMINATIONS AND LIST OF ACTION ITEMS

8.1 Ph.D. Diagnostic/Qualifying Examination

- A. An entering Ph.D. student may be required to take a diagnostic/qualifying examination. This examination, if required by the Major Professor/area of specialization, normally will be taken during the first year of graduate study at ISU.
- B. The purpose of a diagnostic/qualifying examination is to evaluate the student's academic preparation in the topics of critical importance to their Academic Plan. Consequently, examinations may differ depending upon a student's specialization area and research topic.
- C. The student's Major Professor will advise the student as to the necessity for a diagnostic/qualifying examination and arrangements for taking the examination.
- D. Depending upon the outcome of the diagnostic/qualifying examination, a student may be advised to remedy the deficiencies by self-study or taking certain undergraduate courses, for which graduate credits may not be received. These remedial courses must be taken early in a graduate program.

8.2 Master of Science Final Examination

- A. The final examination for M.S. students is administered and evaluated by the student's APC.
- B. The examination is oral, although a written component may be included. Oral examinations usually require a presentation by the student.

- C. The majority of the final exam is directed towards the defense of the thesis or creative component. However, each Major Professor/area of specialization may have further requirements for the final exam.
- D. A complete copy of the thesis or creative component should be delivered to the APC members at least two weeks prior to the final oral examination.
- E. For additional information, refer to the [Graduate College Handbook, Section 7.1.2.](#)

8.3 Doctor of Philosophy Program Examinations

- A. A student undertaking a Ph.D. program must pass a **preliminary examination** prior to being admitted to candidacy for the degree.
 - The preliminary examination will normally include an oral component and a written component taken prior to the oral examination.
 - The written component is coordinated by the Major Professor. This examination is not restricted to the student's research topic and can cover background graduate and undergraduate courses.
 - A graduate student who fails the preliminary examination may be afforded the opportunity to retake it, providing that six months elapse before a re-examination. In this case, the requirements to be satisfied prior to re-examination will be documented and a time limitation will be specified.
 - A student is admitted as a candidate for the Ph.D. degree after successful completion of the preliminary examination.
- B. A Ph.D. candidate must pass a **final examination** to satisfy requirements for the degree.
 - The Ph.D. final examination is devoted primarily to the dissertation defense. The examination is oral and requires a presentation by the student.
 - The APC may recommend re-examination if the student is unsuccessful in passing the examination. In this case, the requirements to be satisfied prior to re-examination will be documented and a time limitation will be specified.
- C. Both preliminary and final examinations are administered and graded by the student's APC.
 - Six months or more must elapse between passing the preliminary oral examination and the date of the final oral examination.
 - For additional information, refer to the [Graduate College Handbook, Section 7.1.2.](#)

8.4 Information and Timing of Required Actions

- A. The CCEE Department and Graduate College procedures, timelines, and forms to be completed are provided in Appendix C.
- B. Exact graduation deadlines for each academic year are announced by Graduate College and posted to the [Graduate College website](#) under **Important Deadlines**.
- C. Students are responsible for meeting all degree policies, deadlines, and requirements. Failure to progress satisfactorily in their degree program is a criterion for dismissal.

9. GRADUATE ASSISTANTSHIP INFORMATION

Every graduate assistantship appointment has expectations of educational duties (in the form of research and teaching) to be performed. The supervisor has a responsibility to evaluate that this obligation is met. A supervisor's judgment that work is unsatisfactory can lead to termination of the appointment following due process as outlined in [Graduate College Handbook, Section 8.9](#). Sometimes the duties of a student's assistantship are not closely tied to a thesis or dissertation project. In such cases, students typically are expected to work on their own research/scholarship in parallel. For additional information on graduate assistantships, refer to the [Graduate College Handbook, Section 3.1](#).

9.1 Letter of Intent (LOI)

- A. A student's Major Professor initiates the LOI for a graduate assistantship appointment by providing specific information regarding:
 - Appointment time (½, ¾, other)

- Stipend amount
- Tuition scholarship percentage
- B. The department's financial team processes the LOI in Workday. When the LOI is ready and approved by the Major Professor, the student will be emailed a notification from the Workday system.
- C. The student should understand the terms of the agreement and ask any questions if needed. If the student agrees to the terms, they can sign the LOI electronically through Workday.
- D. The signed copy of the LOI is saved in Workday. It is recommended that students download a copy for their personal records.
- E. LOI initiation and approval actions are subject to specific payroll and tuition assistance deadlines.

9.2 Scope of Work

- A. For graduate assistantship appointments, work hours depend on the time of appointment:
 - $\frac{1}{2}$ time appointment = 20 hours/week
 - $\frac{1}{4}$ time appointment = 10 hours/week
 - $\frac{3}{4}$ time appointment = 30 hours/week, which can be processed in the following conditions:
 - Any time for U.S. students or international students on CPT
 - Summer only for international students not on CPT
 - Approval for any appointment greater than $\frac{1}{2}$ time is contingent upon proof of reasonable academic progress.
 - Appointments greater than $\frac{3}{4}$ are not allowed.
- B. Coursework and scholarly activities are considered part of a graduate assistant's training, and the combination of assistantship and coursework/scholarly activities equals full time. Course credit loads are as follows:
 - $\frac{1}{2}$ time appointment = 12 credits maximum
 - $\frac{1}{4}$ time appointment = 15 credits maximum
 - Minimum number of credits = 1
- C. International students with a teaching assistantship appointment must complete the Oral English Certification Test (OECT) before they are assigned any teaching duties.
 - Students who do not reach Level 1 (full certification) are required to complete the recommended section(s) of GRST 5400 based on their score.
 - For additional information, refer to the [OECT Website](#).

9.3 Graduate Tuition Scholarship Eligibility

- A. Generally, only students with a cumulative GPA of 3.0 or higher are eligible to hold graduate assistantships and receive the Graduate College tuition scholarship.
 - New students are hired based on their undergraduate GPA.
 - Any student with a GPA below 3.0 as a result of their first term grades will receive a warning letter from the Graduate College and be allowed a grace period of one semester to bring the GPA up to 3.0 or higher.
- B. If a student is placed on academic probation because of a GPA below 3.0, the student will not receive a Graduate College tuition scholarship.

9.4 Stipends

- A. The university establishes a minimum and maximum monthly stipend for $\frac{1}{2}$ time research, teaching, and administrative assistantships. The stipend for a $\frac{1}{2}$ time assistantship is sufficient to cover modest living expenses. Stipends for other fractional appointments are scaled proportionately.
- B. The guidelines and procedures for setting stipends used by the graduate assistant's department are filed in the department office, where they are available to a graduate assistant upon request.
- C. All assistantship monthly stipends are subject to *income tax withholding*.

9.5 Hourly Work in Addition to Assistantship

- A. U.S. citizens and permanent residents who are on graduate assistantships may also work on an hourly basis, provided that:
 - The additional work is approved by the unit awarding the assistantship.
 - The hourly work is not related to the student's educational program.
 - Total hours (assistantship and hourly work) does not exceed 30 hours per week.
- B. Non-immigrant international students must abide by the terms of their visa status and immigration guidelines.
 - For additional information, contact the [International Students and Scholars Office](#).

9.6 Graduate Assistant Employment Benefits

- A. Descriptions below are brief. For full explanation of Employment Benefits, refer to the [Student & Scholar Health Insurance Program](#). Questions can be emailed to isusship@iastate.edu.
- B. Health Insurance
 - Self-only coverage at no cost
 - Eligibility: Students must be on a minimum of ¼ time for at least 3 months fall and spring semesters.
- C. Dental Insurance
 - Self-only coverage
 - Partially subsidized by the university and a monthly premium is deducted from the graduate assistant's paycheck.
 - Eligibility: Students must be on a minimum of ¼ time for at least 3 months fall and spring semesters.
- D. Dependents
 - Lawful spouse or domestic partner and unmarried dependent children under age 26 can be enrolled.
 - They must be added within 30 days of the start date or with a qualifying event.
 - Once added, they will remain until the end of the policy year or when the appointment ends.
 - A portion of the monthly premium will be deducted from the graduate assistant's paycheck when adding coverage for dependents.
- E. Leave of Absence
 - A graduate assistant can request to be absent for illness or personal reasons.
 - Barring unforeseen circumstances, the graduate students on assistantship are expected to plan their personal leave of absence so it does not interfere with or cause neglect of their assistantship duties.
 - Leave of absence arrangements must be made in consultation with the Major Professor to adhere to all grant or other funding source restrictions.
 - Graduate assistant supervisors should be understanding and accommodating when the need for leave occurs but also must ensure that their assistants do not exceed reasonable limits for leave.
 - If a conflict arises between the graduate assistant and supervisor regarding leave of absence use, either party may contact the CCEE DOGE. If the conflict is not resolved in a timely manner, any of the parties may contact the Dean of the Graduate College.
- F. Workers Compensation
 - Injury sustained while performing duties directly related to the graduate assistantship appointment must be reported immediately to University Human Resources (UHR) by the supervisor, instructor and/or department chair using the Incident Reporting System and following the [Report of Injury](#) instructions.
 - Workers compensation may or may not apply to the injury, depending on the situation. Injuries sustained for educational purposes are generally not considered work-related and not usually covered by the worker's compensation system. ISU UHR will provide guidance on claims in consultation with a third-party claims administrator.

9.7 University-Approved Travel

- A. Travel authorization is required by the CCEE Department before any trip that will cause an official workday to be missed or that will use university funds. If a graduate student's appointment is through a unit or center (e.g., Institute for Transportation), additional authorization may be required from that unit or center.
 - The authorization request is required whether travel is hosted or on a grant.
- B. The CCEE Department and InTrans require that the travel authorization form be completed and approved prior to any domestic or foreign travel, including in-state travel.
 - The only exceptions are local travel not requiring reimbursement, or local travel where no standard responsibilities will be missed.
- C. Expenses for out-of-state travel can only be paid if a travel authorization has been approved by the Major Professor. Students should seek approval and agreement with their Major Professor on the availability of funds before travelling.
- D. A travel request should be submitted at least 15 days before the departure, unless there are extenuating circumstances. The student should ask their Major Professor for procedural details when traveling for the first time.
- E. When traveling, student must keep receipts for all expense items that will be included in the reimbursement request.
- F. When a personal car is used, a record of miles traveled must be included.
- G. Graduate students are eligible for some travel grants:
 - [CCEE Graduate Student Travel Scholarship](#) administered by CCEE Graduate Student Council
 - [Professional Advancement Grant \(PAG\)](#) administered by Graduate & Professional Student Senate (GPSS)
 - Pending the availability of funds, graduate students are normally eligible to apply once per academic year. Certain rules and deadlines apply so the guidelines must be reviewed and followed.

10. RESOURCES FOR GRADUATE STUDENTS

10.1 CCEE Office Space

- A. Office space will be provided for teaching and research assistants. The department will accommodate the remaining graduate students as space permits.
- B. Requests for office space should be made through the Major Professor or the focus area graduate coordinator, who will contact the CCEE Department accordingly.
- C. Graduate students located at InTrans will have their desk assignments made by InTrans.

10.2 CCEE Computers

- A. The department maintains and operates networked computer laboratories in Town Engineering Building (e.g., Town 134, 194, and 210) for use by undergraduate and graduate students (any time classes are not being held in them).
- B. Graduate students have a print quota for use of the printers in the computer laboratories and CCEE graduate office spaces.
- C. Most, if not all, office desks for graduate students are equipped with a desktop computer.
- D. Registered graduate students who have a legitimate academic-related need to check out a device from the CCEE Department may make a request by completing the Device Loan Agreement. They are expected to adhere to all the policies and procedures outlined in the referenced agreement.

10.3 CCEE Copy Machines

- A. The CCEE Department's copiers are available with a copier code, and a papercut quota can be used to print to any CCEE printer. The copied/printed materials are expected to be related to academic responsibilities.
- B. A copy code number must be obtained from the graduate student's Major Professor so the costs can be allocated to relevant projects.
- C. Inappropriate use of copy machines (e.g. copying a textbook) is not permitted. It is a state policy that university-owned equipment shall not be used for personal use.

- D. It is important that each person helps maintain security by observing rules for locking equipment room doors.

10.4 CCEE Graduate Student Mailboxes

- A. Mailboxes for graduate students are located in the mail/copy machine room at 397 Town.
- B. The graduate student mailbox bin is on the countertop at the back of the room on the far left side.
- C. Mail is sorted alphabetically by the last name. Each letter of the alphabet has a mail slot.
- D. Graduate students should check their mailboxes at least once a month.

10.5 CCEE Locked Door Access

- A. Students who need door access to laboratories and teaching spaces must contact their major professor or course instructor. Coordination will then be made with the laboratory managers.
- B. Students who need door access to any other building doors, including their assigned offices or Town Engineering Building, must contact the department building manager.
- C. Students who need door access to the InTrans building must contact their major professor at InTrans.

10.6 CCEE Kitchen

- A. Coffee is available in the faculty/staff kitchen at 385 Town.
- B. Payment is by the cup, or the student can join the coffee club by the month or semester.
- C. There are times when tea or cocoa are available, depending on supply.

10.7 ISU Safety Training

- A. Graduate students are responsible for compliance with established safety policies that ensure safety, protection of health, and minimization of the institution's impact on the environment.
- B. All graduate students involved in laboratory or field research are expected to complete Environmental Health & Safety (EH&S) Lab Safety training and any additional lab-specific training needed to complete their research and assistantship duties.
- C. Registration information for safety training is available at the [EH&S website](#).
- D. Students should work closely with their supervisors to identify safety training needs and document the completion of all training.
- E. Graduate students are expected to be familiar with the safety policies, which can be accessed at [Iowa State University Laboratory Safety Manual](#).
- F. Additional information for CCEE safety guidelines, resources, and policies can be found at the [CCEE Safety](#).

10.8 Graduate College Resources

- [Graduate College Handbook](#)
- [Graduate Student Orientation](#)
- [Center for Communication Excellence](#)
- [Student & Major Professor Checklist](#)
- [Career Services](#)
- [Student Forms](#)
- [Degree Deadlines](#)
- [Thesis & Dissertation](#)
- [Graduation Requirements](#)

10.9 Get Involved!

- A. All graduate students are included in the cceegrads@iastate.edu email group and receive the CCEE Department's announcements from this email list.
- B. The CCEE Graduate Student Council is a graduate student organization of the CCEE Department, whose main purpose is to mobilize information and organize events for the CCEE Department's graduate students.

- Email address: cceegsc@iastate.edu
 - Graduate students are automatically members and will be informed of meetings and events by the CCEE Graduate Student Council's leadership.
- C. [ISU Graduate & Professional Student Senate](#)
- D. [ISU College of Engineering Clubs & Organizations](#)
- E. [ISU Student Organizations](#)

11. APPENDIX A: REQUIRED COURSE LISTS FOR SPECIALIZATION AREAS

11.1 Choosing Courses

- A. It is in a student's best interest to discuss course choices with:
- Major Professor and APC
 - Focus area graduate coordinator (if no Major Professor)
 - CCEE DOGE

11.2 Course Offerings

- A. It is a student's responsibility to review course offerings in the ISU Course Catalog and ISU Schedule of Classes to build the Academic Plan. Of most importance:
- How often a course is offered
 - Number of credits
 - Mode of offering and its associated expectations

11.3 Course List Changes

- A. Course lists are subject to change due to:
- ISU catalog offering changes
 - Departmental course offering changes
 - Focus area requirement changes
 - Online course offering changes

11.4 Selection of Courses

- A. **Tables A.1 through A.6** list the courses required by each conventional specialization area within the CCEE Department. The Transdisciplinary Infrastructure Engineering specialization area does not have any required courses.
- B. If a required course is not offered during a student's graduate study at ISU, an alternative course can be included in the Academic Plan (as approved by APC).
- C. All elective courses (within and outside the area of specialization) must be approved by the APC.

Table A.1. Course requirements for the Civil Engineering Materials specialization area

Required for all students when offered	
CE 5810	Geotechnical & Materials Engineering Seminar
Choose 3 courses (minimum) from CE 58X0	
CE 5830	Pavement Analysis and Design
CE 5840	Advanced Design of Concretes
CE 5860	Advanced Asphalt Materials
CE 5870	Advanced Portland Cement Concretes
CE 5890	Pavement Preservation and Rehabilitation
Choose 2 courses (minimum) from CE 56X0	
CE 5600	Fundamentals of Soil Mechanics
CE 5610	Applied Foundation Engineering
CE 5620	Site Evaluations for Civil Engineering Projects
CE 5630	Experimental Methods in Geo-Engineering
CE 5640	Application of Numerical Methods to Geotechnical Design
CE 5650	Fundamentals of Geomaterials Behavior
CE 5670	Geomaterials Stabilization
CE 5680	Dynamics of Soils and Foundations
CE 5690	Ground Improvement

Table A.2. Course requirements for the Construction Engineering & Management specialization area

Required for all students when offered	
CE 5010	Preconstruction Project Engineering & Management
CE 5020	Construction Project Engineering & Management
CE 5030	Construction Finance & Business Management
CE 5940A*	Special Topics: Planning & Scheduling
Choose 1 course (minimum) from Technical Electives	
CE 5050	Design of Construction Systems
CE 5940	Special Topics Series

*Required if undergraduate courses do not include content in planning and scheduling.

The remaining coursework will include:

- Minimum of 3 credit hours from Finance Electives
- Minimum of 3 credit hours from Context Electives

Table A.3. Course requirements for the Environmental Engineering specialization area

Required for all students when offered	
CE 5910	Environmental Engineering Seminar
Required Courses for Water Quality Track	
CE 5200	Environmental Engineering Chemistry
CE 5210	Environmental Biotechnology
CE 5220	Water Pollution Control Processes
CE 5230	Physical-Chemical Treatment Process
Required Courses for Water Quantity Track	
CE 5710	Surface Water Hydrology
CE 5720	Analysis and Modeling of Aquatic Environments
CE 5730	Groundwater Hydrology
CE 5760	Environmental Flows

Table A.4. Course requirements for the Geotechnical Engineering specialization area

Required for all students when offered	
CE 5810	Geotechnical & Materials Engineering Seminar
Choose 4 courses (minimum) from CE 56X0	
CE 5600	Fundamentals of Soil Mechanics
CE 5610	Applied Foundation Engineering
CE 5620	Site Evaluations for Civil Engineering Projects
CE 5630	Experimental Methods in Geo-Engineering
CE 5640	Application of Numerical Methods to Geotechnical Design
CE 5650	Fundamentals of Geomaterials Behavior
CE 5670	Geomaterials Stabilization
CE 5680	Dynamics of Soils and Foundations
CE 5690	Ground Improvement
Choose 2 courses (minimum) from CE 58X0	
CE 5830	Pavement Analysis and Design
CE 5840	Advanced Design of Concretes
CE 5860	Advanced Asphalt Materials
CE 5870	Advanced Portland Cement Concretes
CE 5890	Pavement Preservation and Rehabilitation

Table A.5. Course requirements for the Structural Engineering specialization area

Advanced Structural Analysis Courses	
• Choose 2 courses (minimum) for M.S. & Ph.D. • Choose 3 courses (minimum) for M.Eng.	
CE 5190	Methods for Data-Driven Computational Engineering Research
CE 5320	Structural Analysis II
CE 5410	Dynamic Analysis of Structures
CE 5420	Structural Analysis by Finite Elements
CE 5490	Structural Health Monitoring
Advanced Structural Design Courses	
• Choose 2 courses (minimum) for M.S. & Ph.D. • Choose 3 courses (minimum) for M.Eng.	
CE 5330	Structural Steel Design II
CE 5340	Reinforced Concrete Design II
CE 5350	Prestressed Concrete Structures
CE 5450	Seismic Design
CE 5460	Bridge Design
CE 5480	Building Design

Table A.6. Course requirements for the Transportation Engineering specialization area

Required Courses	
CE 5510	Urban Transportation Planning Models
CE 5530	Traffic Engineering
Statistics: Choose 1 course (minimum)	
CE 5560	Transportation Data Analysis
STAT 5010	Multivariate Statistical models
STAT 5101	Statistical Methods for Data Analysis
STAT 5260	Applied Statistical Modeling
STAT 5750	Introduction to Multivariate Data Analysis
• Choose 3 courses (minimum) for Ph.D. • Choose 2 courses (minimum) for M.S.	
CE 5520	Traffic Safety, Operations, and Maintenance
CE 5570	Transportation System Analysis
CE 5580	Transportation Systems Development and Management
CE 5590	Transportation Infrastructure/Asset Management
CE 6500B	Advanced Topics in Transportation Engineering: Traffic Operations
CE 6500C	Advanced Topics in Transportation Engineering: Data Analysis
CE 6500D	Advanced Topics in Transportation Engineering: Traffic Simulation
CE 5540X	Intelligent Transportation Systems
CE 5550X	Traffic Flow Theories for Existing and Emerging Modes

12. APPENDIX B: MINIMUM UNDERGRADUATE COURSE REQUIREMENTS

Appendix B. Minimum Undergraduate Course Requirements for CCEE Department Graduate Degrees

Minimum Undergraduate Course Requirements	Civil Eng. Materials Geotechnical Eng.	Construction Engineering	Environmental Engineering	Structural Engineering	Transportation Engineering	Transdisciplinary Infrastructure Engineering
Mathematics Calculus (12 semester credits, 3 of the 12 credits can be a numerical analysis course)..... Math 2660...Elementary Differential Equations..... Computer Language..... Statistics.....	Calculus Math 2660 Computer Stat	Calculus Math 2660 Computer Stat	Calculus Math 2660 Computer Stat	Calculus Math 2660 Computer Stat	Calculus Math 2660 Computer Stat	Calculus Math 2660 Computer Stat
Science Chem 1670 or 1770...General Chemistry..... Geol 2010...Geology for Engineers & Envir. Scientists..... Phys 2310...Introduction to Classical Physics I.....	Chem 1670 or 1770 Geol 2010 Phys 2210	Chem 1670 or 1770 Phys 2210	Chem 1670 or 1770 Geol 2010 Phys 2210	 Phys 2210	Chem 1670 or 1770 Geol 2010 Phys 2210	<i>(select one)</i> Chem 1670 or 1770 Geol 2010 Phys 2210
Engineering Mechanics CE 2740...Statics of Engineering..... EM 3240...Mechanics of Materials..... EM 3270...Mechanics of Materials Laboratory..... ME 3450...Dynamics..... ABE 3780...Mechanics of Fluids.....	CE 2740 EM 3240 ME 3450 ABE 3780	CE 2740 EM 3240 EM 3270 ABE 3780	CE 2740 EM 3240 ABE 3780	CE 2740 EM 3240 EM 3270 ME 3450 ABE 3780	CE 2740 EM 3240 ABE 3780	<i>(select six)</i> CE 2740 EM 3240 ME 3450 ABE 3780
Civil Engineering CE 3060...Project Management for Civil Engineers..... CE 3260...Principles of Environmental Engr..... CE 3320...Structural Analysis I..... CE 3330...Structural Steel Design I..... CE 3340...Reinforced Concrete Design I..... CE 3600...Geotechnical Engineering..... CE 3720...Engineering Hydrology and Hydraulics..... CE 3820...Design of Concretes..... CE 3830...Design of Portland Cement Concrete..... CE 4530...Highway Design.....	 CE 3600 CE 3820	CE 3060 CE 3320 CE 3340 CE 3600	 CE 3260 CE 3720	 CE 3320 CE 3330 CE 3340 CE 3600	 CE 3600 CE 3720 CE 3820 CE 4530	 CE 3260 CE 3320 CE 3330 CE 3340 CE 3600 CE 3720 CE 3820 CE 3830 CE 4530
Construction Engineering ConE 3400...Concrete and Steel Construction..... ConE 4220...Construction Cost Estimating..... ConE 4410...Construction Planning, Scheduling, & Control...		ConE 3400 ConE 4220 ConE 4410				ConE 3400 ConE 4220 ConE 4410

13. APPENDIX C: IMPORTANT DEADLINES AND ACTIONS

Appendix C. Information and Timing of Required Actions for CCEE Graduate Students

Requirements	Degree	Timeline	Action	Action Source
Graduate English Requirement	M.Eng. M.S. Ph.D.	Before the start of 1 st semester	English Placement Test (EPT)	EPT Webpage non-native speakers of English
Academic Plan Committee (APC)	M.Eng. M.S.	Before the end of 1 st semester	Academic Plan Committee request	Grad College Deadlines Workday: Submit your Graduate Committee Membership
	Ph.D.	Before the end of 1 st year		
Academic Plan (AP) APC must be completely approved before AP can be submitted.	M.Eng. M.S.	Before the end of the 1st semester. NO LATER than the semester prior to graduation!	Academic Plan Request	Grad College Deadlines Workday: Complete your Academic Plan as a Graduate Student
	Ph.D.	It must be <u>approved</u> 3 months prior to the preliminary oral exam.		
Preliminary Exam After the completion of the majority of coursework	Ph.D. only	Schedule Preliminary Exam (Student arranges the written and oral exam portions with the APC.)	Preliminary Oral Exam Request <i>Submit at least 2 weeks prior to the preliminary exam date</i>	Grad College Deadlines Workday: Request a Graduate Oral Exam
		Complete Preliminary Exam	Major Professor reports the outcome.	Workday: Submit Oral Exam Results
Graduation Preparations	M.S. M.Eng. Ph.D.	By the third week of the graduation semester	Apply for Program Completion	Grad College Deadlines Workday: Apply for Program Completion
Final Oral Exam No final exam for M.Eng.	M.S.	Schedule Final Oral Exam (Upon completion of creative component, thesis, or dissertation, student arranges with the APC.)	Final Oral Exam Request <i>submit at least 3 weeks before final oral exam date</i>	Grad College Deadlines Workday: Submit Oral Exam Results
	Ph.D. (a minimum of 6 months after preliminary exam)	Complete Final Oral Exam	Major Professor reports the outcome.	Grad College Deadlines Workday: Submit Oral Exam Results
Graduation	M.Eng.	At the same time as Apply for Program Completion	Coursework Only Final Check	Grad College Deadlines Workday: Coursework Only Final Check
	M.S. (creative component)	Apply for Program Completion (beginning of the graduation semester) and submit creative component (by the end of the semester), following the deadlines	1) Submit final copy of creative component to Digital Repository 2) Submit Graduate Student Approval Form	Grad College Deadlines ISU Digital Repository Graduate Student Approval Form
	M.S. (thesis)	Apply for Program Completion (beginning of the graduation semester) and submit thesis (by the end of the semester), following the deadlines	1) Open ProQuest Account 2) Submit final copy of thesis to ProQuest 3) Submit Graduate Student Approval Form	Grad College Deadlines ProQuest Account Graduate Student Approval Form
	Ph.D.	Apply for Program Completion (beginning of the graduation semester) and submit dissertation (by the end of the semester), following the deadlines	1) Open ProQuest Account 2) Submit final copy of dissertation to ProQuest 3) Submit Graduate Student Approval Form	Grad College Deadlines ProQuest Account Graduate Student Approval Form

Graduate students are strongly encouraged to check the [ISU Graduate College website](#), including the [Graduate College Handbook](#), on regular basis for the latest information and timeline of actions required for their academic progress and graduation.